CHAPTER 1

Complex numbers: the algebraic point of view

Computing in $\mathbb{C}$, conjugation, binomial formula

Objectives. Compute with the algebraic form; use the conjugate to obtain an inverse or a quotient; solve a linear equation, or a simple equation involving z and $\bar{z}$; expand a power with the binomial formula.

Prerequisites. Algebraic manipulation and standard identities; quadratics; binomial coefficients and Pascal's rule (specialty course); proof by induction.

Fluency

Fluency · No calculator, five minutes.

1. Expand $(2-3x)^2$.
2. Solve in $\mathbb{R}$ the equation $x^2-4x+13=0$.
3. Expand $(a+b)^3$.
4. Give the value of $\binom{5}{2}$.
5. Write $\frac{1}{\sqrt{2}-1}$ without a square root in the denominator.
6. Solve in $\mathbb{R}$ the equation $3x-5=2x+7$.

Solution · Fluency.

1. $4-12x+9x^2$.
2. $\Delta = 16-52 = -36 < 0$: the equation has no real solution. This chapter will give it two.
3. $a^3+3a^2b+3ab^2+b^3$.
4. $\binom{5}{2} = 10$.
5. Multiply by the conjugate expression: $\frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} = \frac{\sqrt{2}+1}{2-1} = \sqrt{2}+1$.
6. $x = 12$.

Introductory activity

Activity 1 · An "impossible" computation that gives the right answer 40 min . In the sixteenth century, a formula was known for solving equations of the form $x^3 = px + q$. It is given here and taken for granted:

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}}.$$

We study the equation $x^3 = 15x + 4$.

1. Check that 4 is a solution of the equation.
2. Compute $\frac{q^2}{4} - \frac{p^3}{27}$ for $p = 15$ and $q = 4$. What becomes of the formula? Can it be applied in $\mathbb{R}$?
3. A calculator of the time decides to carry on as if a number written $\sqrt{-1}$ existed, with the single rule $\sqrt{-1} \times \sqrt{-1} = -1$ and the usual rules of computation for everything else. Expand $\left(2 + \sqrt{-1}\right)^3$ with these rules.
4. Assume that $\sqrt{-121}$ can be written $11\sqrt{-1}$. Show that the first term of the formula can be taken equal to $2 + \sqrt{-1}$, then find in the same way a value for the second term.
5. What does the formula then give? Compare with question 1.
6. Is this computation a proof? Which rules would have to be fixed for it to become one?

Solution · Activity.

1. $4^3 = 64$ and $15 \times 4 + 4 = 64$.
2. $\frac{16}{4} - \frac{3{,}375}{27} = 4 - 125 = -121$. The formula calls for $\sqrt{-121}$, which does not exist in $\mathbb{R}$: it seems useless although the equation has a real solution.
3. $\left(2 + \sqrt{-1}\right)^3 = 8 + 3 \times 4\sqrt{-1} + 3 \times 2 \times (-1) + (-1)\sqrt{-1} = 2 + 11\sqrt{-1}$.
4. The first term is a cube root of $2 + \sqrt{-121} = 2 + 11\sqrt{-1}$; question 3 shows that $2 + \sqrt{-1}$ works. Likewise $\left(2 - \sqrt{-1}\right)^3 = 2 - 11\sqrt{-1}$, so the second term can be taken equal to $2 - \sqrt{-1}$.
5. The sum is $\left(2 + \sqrt{-1}\right) + \left(2 - \sqrt{-1}\right) = 4$: we recover the solution of question 1.
6. No. The symbol $\sqrt{-1}$ has not been defined, and the cube roots were chosen. The course builds a set of numbers, $\mathbb{C}$, with a number i such that $i^2 = -1$ and fixed rules of computation; the notation $\sqrt{-1}$ is avoided there.

Teacher note · Running the activity. Cardano's formula is given and taken for granted: it is not in the syllabus. The point is historical and methodological: a formal computation gives an answer that can be checked, which motivates the construction of $\mathbb{C}$. Point out that cube roots are not unique in $\mathbb{C}$; this is taken up again in chapter 5. The episode is attributed to Bombelli (around 1572).

Course notes

The set of complex numbers

Definition · The set $\mathbb{C}$. We take for granted the existence of a set written $\mathbb{C}$, called the set of *complex numbers*, such that:

1. $\mathbb{C}$ contains $\mathbb{R}$;
2. $\mathbb{C}$ contains a number written i such that $i^2 = -1$;
3. $\mathbb{C}$ has an addition and a multiplication that extend those of $\mathbb{R}$ and follow the same rules of computation (commutativity, associativity, distributivity);
4. every complex number z can be written in a unique way as $z = a + ib$ with a and b real.

The expression $z = a + ib$ is the *algebraic form* of z. The real number a is the *real part* of z, written $\mathrm{Re}(z)$; the real number b is the *imaginary part* of z, written $\mathrm{Im}(z)$. If $b = 0$, z is real; if $a = 0$, z is *purely imaginary*.

Property · Equality of two complex numbers. Two complex numbers are equal if and only if they have the same real part and the same imaginary part. In particular, $a + ib = 0$ with a, b real is equivalent to $a = b = 0$.

Watch out. The imaginary part of $3 - 5i$ is -5, not $-5i$: it is a real number. There is no order compatible with the operations on $\mathbb{C}$: we never write $z < z'$ for non-real complex numbers.

Example 1 · Computations in algebraic form. $(2 + 3i) + (1 - i) = 3 + 2i$ and $(2 + 3i)(1 - i) = 2 - 2i + 3i - 3i^2 = 5 + i$.

The powers of i repeat every four: $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Hence $i^{4k+r} = i^r$ for every integer $k \geq 0$.

Conjugate

Definition · Conjugate. The *conjugate* of $z = a + ib$ (a, b real) is $\overline{z} = a - ib$.

Property · First properties. For every $z = a + ib$:

$$z + \overline{z} = 2\,\mathrm{Re}(z), \qquad z - \overline{z} = 2i\,\mathrm{Im}(z), \qquad z\overline{z} = a^2 + b^2.$$

Hence z is real if and only if $\overline{z} = z$, and purely imaginary if and only if $\overline{z} = -z$. The product $z\overline{z}$ is a nonnegative real number, zero only for $z = 0$.

Proof. $z\overline{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2$. The first two equalities are obtained by adding or subtracting $a + ib$ and $a - ib$. □

Property · Inverse of a nonzero complex number. Every nonzero complex number $z = a + ib$ has an inverse:

$$\frac{1}{z} = \frac{\overline{z}}{z\overline{z}} = \frac{a - ib}{a^2 + b^2}.$$

Method · Writing a quotient in algebraic form. Multiply the numerator and the denominator by the conjugate of the denominator. Example: $\dfrac{1}{3+4i} = \dfrac{3-4i}{9+16} = \dfrac{3}{25} - \dfrac{4}{25}i$.

Property · Conjugate and operations. For all complex numbers z, z' and every natural number n:

$$\overline{z+z'} = \overline{z} + \overline{z'}, \qquad \overline{zz'} = \overline{z}\,\overline{z'}, \qquad \overline{z^n} = \overline{z}^n,$$

and if $z' \neq 0$: $\overline{\left(\frac{1}{z'}\right)} = \frac{1}{\overline{z'}}$ and $\overline{\left(\frac{z}{z'}\right)} = \frac{\overline{z}}{\overline{z'}}$.

Proof. *Product.* Let $z = a + ib$ and $z' = a' + ib'$. Then $zz' = (aa' - bb') + i(ab' + a'b)$, so $\overline{zz'} = (aa' - bb') - i(ab' + a'b)$. On the other hand, $\overline{z}\,\overline{z'} = (a - ib)(a' - ib') = (aa' - bb') - i(ab' + a'b)$. The two results are equal.

Inverse. If $z' \neq 0$, $z' \times \frac{1}{z'} = 1$. By the property of the product, $\overline{z'} \times \overline{\left(\frac{1}{z'}\right)} = \overline{1} = 1$, so $\overline{\left(\frac{1}{z'}\right)}$ is the inverse of $\overline{z'}$. The quotient follows by writing $\frac{z}{z'} = z \times \frac{1}{z'}$.

Power. By induction on n. For $n = 0$: $\overline{z^0} = \overline{1} = 1 = \overline{z}^0$. If $\overline{z^n} = \overline{z}^n$ for some integer n, then $\overline{z^{n+1}} = \overline{z^n \times z} = \overline{z^n} \times \overline{z} = \overline{z}^n \times \overline{z} = \overline{z}^{n+1}$. □

Equations

Method · Linear equation. If $a \neq 0$, the equation $az = b$ has a unique solution $z = \frac{b}{a}$, which we write in algebraic form. Example: $(1+i)z = 3 - i$ gives $z = \frac{(3-i)(1-i)}{2} = \frac{2-4i}{2} = 1 - 2i$.

Method · Equation involving z and $\overline{z}$. Let $z = x + iy$ with x, y real, write each side in algebraic form, then identify real parts and imaginary parts. This gives a system of two real equations.

Example 2 · Solving $2z + i\overline{z} = 3 - 3i$. With $z = x + iy$: $2x + 2iy + i(x - iy) = (2x + y) + i(x + 2y)$. Identification gives $2x + y = 3$ and $x + 2y = -3$, hence $x = 3$ and $y = -3$. The solution is $z = 3 - 3i$. Check: $2(3 - 3i) + i(3 + 3i) = 6 - 6i + 3i - 3 = 3 - 3i$.

Binomial formula

Property · Binomial formula. For all complex numbers a, b and every natural number n:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

Proof. By induction on n. For $n = 0$, both sides equal 1. Assume the formula holds for some integer n. Then

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^{n} \binom{n}{k} a^{k+1} b^{n-k} + \sum_{k=0}^{n} \binom{n}{k} a^k b^{n+1-k}.$$

In the first sum, set $j = k + 1$: it becomes $\sum_{j=1}^{n+1} \binom{n}{j-1} a^j b^{n+1-j}$. Grouping the terms with the same exponent of a and using Pascal's rule $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$ for $1 \leq j \leq n$, we obtain

$$(a+b)^{n+1} = b^{n+1} + \sum_{j=1}^{n} \binom{n+1}{j} a^j b^{n+1-j} + a^{n+1} = \sum_{j=0}^{n+1} \binom{n+1}{j} a^j b^{n+1-j}.$$

The formula holds for $n + 1$, hence for every natural number n. □

Example 3 · A power of $1+i$. $(1+i)^4 = 1+4i+6i^2+4i^3+i^4 = 1+4i-6-4i+1 = -4$. One can also write $(1+i)^2 = 2i$, then $(1+i)^4 = (2i)^2 = -4$.

Key point · Key points. Unique algebraic form; $i^2 = -1$; $z\bar{z} = a^2+b^2$; for a quotient, multiply by the conjugate of the denominator; for an equation in z and $\bar{z}$, set $z = x+iy$ and identify.

Exercises

Exercise 1.1 · Algebraic form. Write in algebraic form:

1. $(3-2i)+(-1+5i)$;
2. $(3-2i)(-1+5i)$;
3. $(1-2i)^2$;
4. i^{2026}.

Solution.

1. $2+3i$.
2. $-3+15i+2i-10i^2=-3+17i+10=7+17i$.
3. $1-4i+4i^2=-3-4i$.
4. $2{,}026=4\times 506+2$, so $i^{2026}=i^2=-1$.

Exercise 1.2 · Inverses and quotients. Write in algebraic form: $\dfrac{1}{1+i}$; $\dfrac{2+i}{1-2i}$; $\dfrac{3+i}{2-i}$.

Solution. $\dfrac{1}{1+i}=\dfrac{1-i}{2}=\dfrac{1}{2}-\dfrac{1}{2}i$. Next $\dfrac{(2+i)(1+2i)}{5}=\dfrac{2+4i+i-2}{5}=i$. Finally $\dfrac{(3+i)(2+i)}{5}=\dfrac{6+3i+2i-1}{5}=1+i$.

Exercise 1.3 · Equations. Solve in $\mathbb{C}$:

1. $(1+i)z=3-i$;
2. $2z-3\overline{z}=1+10i$;
3. $z+2\overline{z}=6-i$.

Solution.

1. $z=\dfrac{3-i}{1+i}=\dfrac{(3-i)(1-i)}{2}=\dfrac{2-4i}{2}=1-2i$.
2. With $z=x+iy$: $2x+2iy-3x+3iy=-x+5iy$. Identification gives $-x=1$ and $5y=10$, that is $z=-1+2i$.
3. $x+iy+2x-2iy=3x-iy$, so $3x=6$ and $-y=-1$: $z=2+i$.

Exercise 1.4 · Real part and imaginary part of an expression. For $z=x+iy$ (x, y real), let $Z=z^2-2\overline{z}$.

1. Express $\mathrm{Re}(Z)$ and $\mathrm{Im}(Z)$ in terms of x and y.
2. Determine the complex numbers z for which Z is real.

Solution.

1. $z^2 = x^2 - y^2 + 2ixy$ and $-2\overline{z} = -2x + 2iy$. So $\mathrm{Re}(Z) = x^2 - y^2 - 2x$ and $\mathrm{Im}(Z) = 2xy + 2y = 2y(x+1)$.
2. Z is real if and only if $2y(x+1) = 0$, that is $y = 0$ or $x = -1$: z is real, or z has real part -1.

Exercise 1.5 · Binomial formula.

1. Using the binomial formula, write $(1+i)^5$ in algebraic form.
2. Deduce $(1+i)^{10}$.
3. Recover these two results from $(1+i)^2$.

Solution.

1. $(1+i)^5 = 1 + 5i + 10i^2 + 10i^3 + 5i^4 + i^5 = 1 + 5i - 10 - 10i + 5 + i = -4 - 4i$.
2. $(1+i)^{10} = (-4-4i)^2 = 16(1+i)^2 = 16 \times 2i = 32i$.
3. $(1+i)^2 = 2i$, so $(1+i)^4 = -4$, $(1+i)^5 = -4(1+i) = -4 - 4i$ and $(1+i)^{10} = (2i)^5 = 32i^5 = 32i$.

Exercise 1.6 · A purely imaginary quotient. Let z be a complex number such that $z\overline{z} = 1$ and $z \neq 1$. Let $w = \dfrac{1+z}{1-z}$.

1. Explain why $\overline{z} = \dfrac{1}{z}$.
2. Compute $\overline{w}$ and prove that w is purely imaginary.
3. Check the result for $z = i$.

Solution.

1. $z\overline{z} = 1$ forces $z \neq 0$, and $\overline{z} = \frac{1}{z}$.
2. By the properties of the conjugate, $\overline{w} = \dfrac{1+\overline{z}}{1-\overline{z}} = \dfrac{1+\frac{1}{z}}{1-\frac{1}{z}} = \dfrac{z+1}{z-1} = -w$. So w is purely imaginary.
3. $w = \dfrac{1+i}{1-i} = \dfrac{(1+i)^2}{2} = i$, which is indeed purely imaginary.

Exercise 1.7 · Sum of the powers of i.

1. Prove that, for every complex number $q \neq 1$ and every natural number n, $1 + q + \cdots + q^n = \dfrac{1 - q^{n+1}}{1 - q}$.
2. Deduce $S = 1 + i + i^2 + \cdots + i^{2026}$.
3. Recover this result by grouping the terms in blocks of four.

Solution.

1. If $S_n = 1 + q + \cdots + q^n$, then $S_n - qS_n = 1 - q^{n+1}$ (the other terms cancel). Since $1 - q \neq 0$, we divide. The proof only uses the rules of computation, which hold in $\mathbb{C}$.
2. $2{,}027 = 4 \times 506 + 3$, so $i^{2027} = i^3 = -i$ and $S = \dfrac{1 + i}{1 - i} = \dfrac{(1 + i)^2}{2} = i$.
3. Four consecutive powers add up to $i^{4k}(1 + i - 1 - i) = 0$. The 2,027 terms form 506 blocks with zero sum, and what remains is $i^{2024} + i^{2025} + i^{2026} = 1 + i - 1 = i$.

Problem

Problem · Square roots of a complex number. We look for the complex numbers z such that $z^2 = 3 + 4i$. Let $z = x + iy$, with x and y real.

1. Prove that $z^2 = 3 + 4i$ is equivalent to the system $x^2 - y^2 = 3$ and $2xy = 4$.
2. Prove that if $z^2 = 3 + 4i$, then $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + (2xy)^2$, and deduce that $x^2 + y^2 = 5$.
3. Deduce the two solutions of the equation, then check them.
4. By the same method, determine the complex numbers whose square is $-5 + 12i$.
5. Consider the equation $z^2 - 3z + 3 - i = 0$.
   (a) Prove that it can be written $\left(z - \frac{3}{2}\right)^2 = \frac{-3+4i}{4}$.
   (b) Determine the complex numbers w such that $w^2 = -3 + 4i$.
   (c) Deduce the solutions of the equation and check their sum and their product.

Solution · Problem.

1. $z^2 = x^2 - y^2 + 2ixy$. By uniqueness of the algebraic form, $z^2 = 3 + 4i$ is equivalent to $x^2 - y^2 = 3$ and $2xy = 4$.
2. The identity $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + 4x^2y^2$ is checked by expanding. It gives $\left(x^2 + y^2\right)^2 = 9 + 16 = 25$, and since $x^2 + y^2 \geq 0$, $x^2 + y^2 = 5$.
3. Adding and then subtracting $x^2 - y^2 = 3$: $x^2 = 4$ and $y^2 = 1$. Since $xy = 2 > 0$, x and y have the same sign: $z = 2 + i$ or $z = -2 - i$. Check: $(2 + i)^2 = 4 + 4i - 1 = 3 + 4i$, and $(-2 - i)^2 = (2 + i)^2$.
4. $x^2 - y^2 = -5$, $xy = 6$ and $x^2 + y^2 = \sqrt{25 + 144} = 13$. So $x^2 = 4$, $y^2 = 9$, with x and y of the same sign: $z = 2 + 3i$ or $z = -2 - 3i$. Check: $(2 + 3i)^2 = 4 + 12i - 9 = -5 + 12i$.
5. (a) $\left(z - \frac{3}{2}\right)^2 = z^2 - 3z + \frac{9}{4}$, so the equation can be written $\left(z - \frac{3}{2}\right)^2 = \frac{9}{4} - 3 + i = \frac{-3+4i}{4}$.
   (b) $x^2 - y^2 = -3$, $xy = 2$, $x^2 + y^2 = 5$: $x^2 = 1$, $y^2 = 4$, hence $w = 1 + 2i$ or $w = -1 - 2i$.
   (c) $z - \frac{3}{2} = \pm\frac{1+2i}{2}$, hence $z_1 = 2 + i$ and $z_2 = 1 - i$. Their sum is 3 and their product is $(2 + i)(1 - i) = 2 - 2i + i + 1 = 3 - i$, as the coefficients of the equation predict.

Teacher note · Status of the problem. Complex square roots and quadratic equations with complex coefficients are "possible problems" of the syllabus, not required. Question 2 avoids the modulus, introduced in the next chapter: one can come back to it by noting that $x^2 + y^2 = |z|^2 = |z^2|$.